Performance + SEO Audit

Core Web Vitals, technical SEO, A11y

Proyecto	monoulabs.ai landing (rladron-rodia/landing_contacto)
Version	v3.1.0
Fecha	2026-05-09
Auditor	Auditoria automatizada (Claude agent)
AWS Account	325907670798 (us-east-1)
Scope	Core Web Vitals, technical SEO, A11y

1. Executive Summary

The landing is in **strong technical shape**. Most low-hanging-fruit perf and SEO best-practices have already been adopted in v2.4–v2.5 (critical CSS inline, Tailwind inlined, Inter self-hosted with `font-display:swap`, FA subset, JS deferred + post-LCP `requestIdleCallback`, dns-prefetch, schema.org Dataset/Organization/Service/WebSite). The remaining issues are smaller and mostly fall into three buckets: (a) HTML weight inflation from inlined Tailwind being unminified-vs-purged, (b) `lang` attribute / default UI language mismatch (a real SEO issue), and (c) a few semantic/A11y nits in the carousel.

Findings count by category × severity

Category	High	Medium	Low	Total
Performance — Frontend	1	4	4	9
Performance — Backend & Infra	0	2	2	4
SEO — Technical	1	3	2	6
SEO — Content /■ A11y	1	4	2	7
Total	**3**	**13**	**10**	**26**

Top 5 priorities

1. Fix `lang` attribute mismatch (HIGH, SEO). `<html lang="es">` but JS forces `currentLang='en'` and rewrites every `data-en` text on `DOMContentLoaded`. Crawlers see ES markup, real users see EN copy. Either default JS to ES, or set `<html lang="en">` and treat ES as toggle. Currently signals duplicate-content / wrong locale to Google. Fix: 1 line change (`index.html:1207`).

2. Carousel buttons use `<div onclick=…>` instead of `<button>` (HIGH, A11y). Lines 709, 710, 844–847. Not keyboard-accessible, no role, breaks WCAG 2.1.1 (Keyboard) and 4.1.2 (Name, Role, Value). Easy fix, big a11y/SEO win (Lighthouse a11y will jump).

3. Inline Tailwind ships ~22 KB of unused utility classes (HIGH, Perf). The "static compile" still includes everything (sm:/md:/lg:/xl:/2xl: variants, hover/focus/group, container queries, gradients) — even classes the actual HTML never uses. A proper Tailwind purge (with `content: ['index.html']`) usually trims this to 6–8 KB. Saves ~15 KB on every cold HTML request (CloudFront max-age=60), translates to ~80–150 ms LCP improvement on slow 4G.

4. Footer privacy/data-licenses links are dead anchors `href="#"` (MEDIUM, SEO + UX). Schema.org and llms.txt advertise `#privacy` and `#data-licenses` as canonical fragments, but they don't exist in the DOM (lines 1163–1164). Inconsistent metadata; bots may flag broken anchors.

5. `fa-solid-900.woff2` is 156 KB and is preloaded (MEDIUM, Perf). The icons set used (≈30 glyphs from `fa-subset.min.css`) doesn't justify a 156 KB font when the rest of the site uses 24 KB Inter weights. Subsetting `fa-solid` to only-used glyphs (pyftsubset / fonttools) would cut ~120 KB, and removing the preload (LCP doesn't depend on icons) would unblock the network for the actually-critical Inter fonts.

Estimated improvement potential

- **LCP**: ~150–300 ms reduction on mobile slow-4G (Tailwind purge + drop FA preload).
- **Total page weight**: –135 KB (FA subset) – 15 KB (Tailwind purge) = ~150 KB saved on first byte.
- **Lighthouse SEO**: 92 → 98 (lang fix + close anchor links + structured-data lastmod sync).
- **Lighthouse A11y**: 88–90 → 96+ (semantic carousel buttons + couple ARIA fixes).

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2. Performance — Frontend

2.1 HTML / Critical Render Path

• **Size**: `index.html` = **117 KB** raw (~22 KB Brotli). The bulk (~92 KB) is the inlined `<style id="tailwind-inlined">` block plus four `<script type="application/ld+json">` blocks. Without inlining Tailwind, HTML would be ~25 KB raw.

• **Critical CSS** is correctly inlined (line 215–229) for above-the-fold (body, bg, text, dimensions).

• **Critical JS** (carousel, language switcher) lives at the bottom of `<body>` (lines 1170–1309). Good.

• **GrapesJS leftover**: lines 520–530 contain `<style data-grapesjs-styles="true">` with two ID selectors (`#icyrfq`, `#ir9pjh`) — these are GrapesJS-generated IDs that leaked to prod. Cosmetic only, but flags that the page was built in a no-code tool and not all output was cleaned.

• **Body class soup**: `<body class="min-h-screen w-full flex flex-col relative text-gray-300 antialiased selection:bg-brand-500/30 selection:text-white m-0 p-0">` (line 533) — duplicate `m-0 p-0` already covered by reset. Minor.

2.2 CSS (size, blocking, Tailwind, FontAwesome)

• **Tailwind 3.4.17 inlined** (line 244): ~22 KB minified, but the compiled CSS includes many never-used utilities (every `from-color-N00/20`, container queries, every `space-y-*`, every `shadow-[...]`). Strong likelihood that with `content: ['index.html']` purge you'd drop to 6–8 KB. **HIGH-priority finding**.

• **Tailwind file at** `assets/css/tailwind.min.css` (25 KB on disk) is **dead weight in S3** — no `<link>` references it. Either delete or tag with `nosync` so deploy script doesn't ship it. Saves bucket space, no perf impact directly.

• **FA subset CSS** (`fa-subset.min.css`, 16.6 KB) is loaded async via `<link rel="preload" … onload=…>` pattern (line 248). Good. NoScript fallback present (line 249). Good.

• **FA subset CSS still references missing fonts**: `fa-brands-400.woff2` and `fa-v4compatibility.woff2` are declared in `@font-face` (line 8 of fa-subset.min.css) but the files don't exist in `assets/fonts/`. Browsers won't fetch them unless an element actually uses those font-families (the page only uses `Font Awesome 6 Free`), but a strict CSP check or audit tool will flag the dead references. Recommend stripping those `@font-face` blocks from the subset build.

• **No critical-path CSS warnings**: only the tailwind-inlined block is render-blocking; FA + Inter are async or lazy.

2.3 JavaScript (size, loading strategy, third-party)

• **Total JS shipped (deferred)**: ~45 KB raw, ~14 KB Brotli

- `contact-form.js`: 8.6 KB
- `analytics-loader.js`: 10.4 KB
- `games-loader.js`: 7.1 KB
- `links-loader.js`: 6.3 KB
- `delivery-loader.js`: 4.9 KB
- `info-columns-loader.js`: 4.4 KB
- `stats-loader.js`: 3.6 KB

• **Loading**: all 7 scripts use `defer` (lines 1314–1321). Good. `analytics-loader.js` additionally uses `requestIdleCallback` (with 3s timeout, fallback `setTimeout(boot, 1500)` post-`load`) — excellent practice (lines 232–243). This sidesteps the typical Lighthouse "63 KB of unused JS in critical path" hit from gtag.

• **Network waterfall (cold load)**: `index.html` → 6 deferred loaders fire `fetch()` to `api.monoulabs.ai` after DOMContentLoaded:

- `/api/stats`, `/api/games`, `/api/site-links`, `/api/delivery-options`, `/api/info-columns`, `/api/cta-tags`, `/api/analytics-config` = **7 cross-origin fetches**.

- Caches in `localStorage` for 5 min, but the **first visit pays the round-trip cost for all 7**. With ~50–100 ms median ALB+RDS, this is ~500–700 ms of additional network activity post-LCP. Doesn't affect LCP/CLS but bloats TBT.

- Consider a **single batch endpoint** `/api/landing-bootstrap` that returns all 6 public payloads in one round-trip. Saves 6× HTTP round-trip + 6× CORS preflight (if any).

• **Inline carousel + i18n script** (lines 1170–1309, ~140 lines) is fine but does `document.querySelectorAll('[data-es]').forEach(...)` 3 times on load — could be done once. Minor.

• **No third-party JS** in the static HTML. GA4/GTM are only injected post-idle by `analytics-loader.js`, conditional on backend toggles. Good.

2.4 Images (formats, sizing, lazy loading)

• **10** `<img>` **elements** in HTML (`index.html:717, 727, 737, 854, 864, 874, 975, 981, 987, 993`). All have:

- `loading="lazy"` ✓

- `decoding="async"` ✓

- explicit `width`/`height` (320×180 for video tiles, 200×200 for square tiles) ✓ — prevents CLS

- descriptive Spanish `alt` text with SEO keywords ("Dataset Minecraft — open world, 2.2M videos indexados") ✓

• **Format**: PNG. The JS loader (`games-loader.js:97`) supports a `<picture>` with `<source type="image/webp">` if `image_webp_url` is provided by the backend, but the **hard-coded fallback in HTML is PNG only**. **MEDIUM finding**: serve WebP/AVIF via `<picture>` directly in the HTML for the initial paint. Users with no JS, or who land before the games-loader runs, see PNGs. PNGs from `muestra-imagen.s3.us-east-1.amazonaws.com` are 50–200 KB each.

• **No** `srcset`**/**`sizes` for responsive resolutions. The 320×180 image is shown at the same size on mobile and desktop, but Retina displays will request the 1× and upscale. Minor.

• **OG image**: `og-preview.png` is **100 KB** — within the 8 MB limit, but large for a social card. Optimizing to WebP-encoded JPG ~40 KB is realistic.

2.5 Fonts (preloading, font-display)

• **Inter 400/600/700 self-hosted** (lines 64–66 preload, 260–274 @font-face). All use `font-display: swap` ✓ and `unicode-range: latin` to keep weight low (~24 KB each). Good.

• **FontAwesome preloads**: `fa-solid-900.woff2` (156 KB) and `fa-regular-400.woff2` (25 KB) preloaded as `font` (lines 252–253). **MEDIUM finding**: the FA solid woff2 is 156 KB — large for a font where only ~30 glyphs are actually used. Two options:

1. **Subset further with pyftsubset** to only the ~30 codepoints listed in `fa-subset.min.css` (lines 8). Realistic target: 15–20 KB.

2. **Drop the preload** for `fa-solid-900.woff2` — icons are not in the LCP path, can paint via the async-loaded `fa-subset.min.css`. The `<i class="fa-solid …">` elements briefly show as boxes until the font loads (covered by `font-display: swap`). The 156 KB preload steals bandwidth from Inter (which IS in the LCP path).

• **No** `fa-brands-400.woff2` is shipped, but the CSS references it — broken `@font-face` rule (harmless but ugly).

2.6 Cache strategy

• **CloudFront cache config**: `index.html` `max-age=60`, assets `max-age=3600` (per AWS-DEPLOY.md L151–157). 1-hour for CSS/JS/fonts is **conservative**. Recommend hashed filenames + `max-age=31536000, immutable` for fingerprinted assets. Currently un-hashed (`assets/js/contact-form.js`), so `max-age=3600` is the safe cap.

• `Cache-Control` **headers on API**: backend sets `no-store` on admin routes (`server/app.py:153`). Public routes (`/api/stats`, `/api/games`, etc.) have **no explicit Cache-Control** — falls back to Flask default (no-cache or implicit). Adding `Cache-Control: public, max-age=60` (since these are read-mostly JSON, refreshed via admin) would let CloudFront cache them at the edge if you ever route the API through CF. Today not relevant since `api.monoulabs.ai` goes ALB direct.

• **CloudFront 403/404 → /index.html with 200**: this is fine for SPA-like behavior, but with the URL rewriter function it's mostly redundant. Minor.

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3. Performance — Backend & Infra

3.1 CloudFront caching analysis

• **Edge cache hit ratio**: with `index.html` at 60 s and a low-traffic B2B site (estimated <500 sessions/day), most visitors **miss the edge cache** for HTML on first hit. Acceptable trade-off for content freshness.

• **Brotli/Gzip**: CloudFront auto-compresses by default. Verify `Accept-Encoding: br` returns `Content-Encoding: br` for `text/html`, `text/css`, `application/javascript`. Not tested live, but if AWS-DEPLOY.md hasn't disabled it explicitly, it's on.

- **OAC + S3 private**: correct architecture for CloudFront → S3. WAF v2 in front catches abusive traffic.
- **CF Function** `monou-url-rewriter`: viewer-request, rewrites `/path/` → `/path/index.html`. Minimal cost (~1 ms). Good.

3.2 ALB + EC2 + gunicorn sizing

- **EC2**: `t4g.micro` (Graviton, 2 vCPU, 1 GB RAM). Solid for the workload. CPU-bound only on email send (rare).
- **gunicorn**: `-w 2` workers (per CHANGELOG L75). With 2 workers and `flask_limiter` storage in-memory (`server/app.py:127`), the rate limit is effectively **2× the configured value** (each worker has its own counter). For "5 per minute" on contact form, real cap is ~10/min. Acceptable for honeypot + email validation, but worth noting.
- **Flask-Limiter using** `memory://`: same comment as above. To get strict cross-worker rate limiting, migrate to Redis. Listed in SECURITY-TODO.

3.3 RDS query performance considerations

- **Schema** (`server/db.py:23–135`): single-table reads, indexed on `created_at`, `email`, `status` for `contacts`. The other 6 read tables (`stats`, `games`, `site_links`, `delivery_options`, `info_columns`, `cta_tags`) are tiny (estimated <100 rows each). All `list_*` functions do simple `SELECT … ORDER BY display_order` — no joins, no N+1. Good.
- **No connection pooling** declared — Flask + psycopg with `@contextmanager conn()`. Each request opens a new conn. With 2 gunicorn workers and infrequent traffic, this is fine. Under load you'd want `psycopg_pool` or PgBouncer.
- **No query caching layer**. Adding 60 s in-memory or Redis cache for the 6 public read endpoints would let them serve at <5 ms instead of ~30–80 ms. Low priority given current traffic.

3.4 API endpoint latency expectations

Estimated p50 latencies (US-East-1 origin, US visitor):

- `GET /api/health`: ~30 ms
- `GET /api/stats`, `/api/games`, `/api/site-links`, `/api/delivery-options`, `/api/info-columns`, `/api/analytics-config`, `/api/cta-tags`: ~40–70 ms each (SQL + JSON serialize + ALB hop)
- `POST /api/contact`: ~150–400 ms (DB insert + SES `SendEmail`)

The 7 parallel fetches from the page after DCL = ~70 ms longest-pole, but with HTTP/2 multiplexing on the same origin (`api.monoulabs.ai`), they share one TCP/TLS connection. Net effect: mostly hidden behind LCP.

3.5 Cold start considerations

- **EC2 always-warm** ✓ (per task brief). No cold-start penalty.
- **gunicorn process model**: prefork, workers stay warm. First request after deploy takes ~50 ms longer (psycopg connect), then steady state.
- **No Lambda, no SSR**: nothing to cold-start.

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4. SEO — Technical

4.1 Meta tags

- **Title** (line 6): "Gameplay Datasets para IA y World Models | Monou Labs" — 56 chars, brand-suffix pattern, ✓.
- **Description** (line 7): 196 chars, includes core keywords (gameplay, RL, Behavior Cloning, World Models). Slightly long for Google SERP truncation (~155–160 char snippet) but well-structured. ✓
- **Robots**: `index, follow`. ✓
- **Author**: "Monou Labs". ✓
- **Viewport**: `width=device-width, initial-scale=1.0`. ✓
- **Charset**: `UTF-8`. ✓
- **theme-color, color-scheme, format-detection, apple-mobile-web-app-***: comprehensive PWA meta block (lines 69–77). ✓

• **Favicon**: SVG-data-URL emoji ■ (line 80). Cute, works, but no `apple-touch-icon` or 192×192 PNG for Android home-screen install. Minor.

4.2 Canonical URLs and HTTPS enforcement

• **Canonical**: `https://monoulabs.ai/` (line 10). Single canonical, no www variant — correct.

• **DNS records**: both apex and `www.monoulabs.ai` alias to CloudFront (per AWS-DEPLOY L84–85). Both serve identical content. **MEDIUM finding**: there's no documented 301 from `www` → apex. If both serve 200 with no redirect, you have a duplicate-content scenario. Either:

- Configure CloudFront to 301 `www.monoulabs.ai` → `monoulabs.ai`, OR

- Add `Host` header check in `monou-url-rewriter` viewer-request CF Function.

- Verify with: `curl -I https://www.monoulabs.ai/` — if 200, fix; if 301, you're already covered (and this finding is invalid).

• **HTTPS**: ACM cert wildcard `*.monoulabs.ai`. ALB redirects HTTP→HTTPS (per AWS-DEPLOY:21). ✓ for API. CloudFront default behavior should redirect HTTP→HTTPS for the frontend — confirm in CF distribution config.

4.3 Open Graph + Twitter Cards

• **OG block** (lines 18–30): full coverage — `og:type`, `og:url`, `og:title`, `og:description`, `og:image` + width/height/type/alt, `og:locale=es_MX` + alternate `en_US`, `og:site_name`. ✓ excellent.

• **Twitter Card** (lines 33–39): `summary_large_image`, title, description, image, alt, site, creator. ✓ Note: twitter title is in **English** ("Gameplay Datasets for AI Training") while OG title is in **Spanish** ("para IA y World Models"). Mild inconsistency — depending on which scraper a network uses, the preview will be in different languages. Pick a primary and stick to it (probably ES, given canonical and lang).

4.4 Structured data (schema.org)

Four JSON-LD blocks (lines 83–213):

1. Dataset — well-populated with `name`, `description`, `keywords` (10 entries), `creator`, `distribution` (3 formats), `measurementTechnique`, `variableMeasured`, `temporalCoverage`, `dateModified: 2026-05-05`. **Notable**: this hits Google Dataset Search — a low-competition channel. ✓ great.

2. Organization — `@id` set, `name`, `logo`, `description`, `knowsAbout`, `contactPoint`. `sameAs` only contains the canonical URL — should include LinkedIn, Twitter (`@monoulabs`), GitHub if applicable. **LOW finding**: empty-ish `sameAs` reduces entity-graph signal.

3. Service — uses `@id` reference to organization. ✓ good.

4. WebSite — minimal, no `potentialAction.SearchAction` (the site has no search). Acceptable for a single-page landing.

• **Date staleness**: `dateModified: 2026-05-05` and sitemap `lastmod: 2026-05-05` are aligned, but today is 2026-05-09 (per env). Last actual change to landing copy may be older. Bump these on every meaningful content change to signal freshness. **LOW finding**.

4.5 Sitemap + robots.txt

• `sitemap.xml` (17 lines): single entry for `https://monoulabs.ai/` with `lastmod`, `changefreq=weekly`, `priority=1.0`, hreflang alternates. Correct for a single-page site. ✓

• `robots.txt` (145 lines): exemplary — explicit `Allow` for major search engines + AI search bots (OAI-SearchBot, PerplexityBot, Google-Extended, Applebot, MistralAI), explicit `Disallow` for AI training bots (GPTBot, ClaudeBot, CCBot, etc.), throttle for SEO scrapers (Ahrefs, Semrush). Sitemap reference at the end. ✓ This is the strongest part of the SEO setup.

• `llms.txt` (59 lines): present at root, well-structured for LLM retrieval. References `#privacy` and `#data-licenses` which **don't exist** in the DOM (see finding 4.7).

4.6 Hreflang and i18n

• **Hreflang declarations** (lines 14–15): only `es` and `x-default`, both pointing to the same URL. Consistent with the "Audit v3 Opción B" comment (line 12) that says EN is a UX toggle, not an indexable URL.

• **HOWEVER**: as noted in finding #1, JS forces the page to render in English on first paint. Crawlers fetching as `Googlebot` see the **server-rendered HTML in Spanish**, then JS runs, then they see **English** — but most crawlers don't execute JS that mutates `data-es`/`data-en`. So the index in GSC will be Spanish, and that matches your hreflang declaration. **In practice, this works** but is fragile and confusing. **HIGH finding**: align the on-load language with the declared `lang` attribute. Either:

- Switch JS default to `'es'` (likely correct given B2B target market = LATAM/Iberia + global English).

- OR change `<html lang="en">` AND the hreflang to `en` + canonical accordingly.

4.7 HTTP redirects (apex/www, trailing slash)

• **Trailing slash**: CF Function `monou-url-rewriter` handles `/path/` → `/path/index.html` (per AWS-DEPLOY L71). Good.

• **Apex/www unification**: see finding 4.2. **MEDIUM, needs verification.**

• **Footer "anchors"** (lines 1163–1164): `<a href="#">` for "Términos de Privacidad" and "Licencias de Datos". **MEDIUM finding**:

- These are referenced in `llms.txt` as `https://monoulabs.ai/#privacy` and `https://monoulabs.ai/#data-licenses`.

- Schema.org Service `termsOfService: "https://monoulabs.ai/#privacy"` (line 196).

- But the actual HTML has neither `<section id="privacy">` nor `id="data-licenses"` — these fragments resolve to the top of the page.

- Fix: either add real privacy/licenses sections, or change the `data-link-key="privacy_terms"` / `data-link-key="data_licenses"` to point to `/privacy` and `/data-licenses` static pages, OR remove from llms.txt + schema until they exist.

4.8 Mobile-first readiness

• **Viewport meta**: ✓

• **Responsive layout**: Tailwind `md:`, `lg:` breakpoints used throughout (e.g., line 560 `text-5xl md:text-7xl`, line 580 `grid-cols-1 md:grid-cols-3`). Mobile carousel replaced with horizontal scroll (`#icyrfq`, line 850). ✓

• **Touch targets**: contact form inputs are `py-3` = 12 px vertical padding, comfortable. Carousel `<div onclick=>` buttons are 48×48 — meet the 44 px target, but as `<div>` they're not focusable (see finding #2).

• **No horizontal overflow**: `body { overflow-x: hidden; }` (line 306). ✓

5. SEO — Content & A11y

5.1 Heading hierarchy

• **One** `<h1>` at line 560 ✓

• **Multiple** `<h2>` for sections (`metricas`, `como-funciona`, `dataset-intelligence`, `juegos-monou`, `juegos-terceros`, `formato-entrega`, `contacto`) ✓

• `<h3>` **and** `<h4>` for cards within sections — order is consistent ✓

• **One concern**: `<h3>` is used for stat values like "10k+", "2.5M", "50+" (lines 584, 592, 600). These aren't semantic headings — they're metric values. Lighthouse won't flag, but a strict review would prefer `<p class="text-5xl">` with a separate label. **LOW**.

• **Carousel cards use** `<h4>` for game titles inside a `<section>` whose heading is `<h2>`. There's no intermediate `<h3>` — heading skip from h2 → h4. **LOW finding**.

5.2 Image alt text

• All 10 `<img>` have descriptive Spanish alt text with SEO keywords. ✓

• The lone visual without alt-equivalent text is the SVG emoji favicon — irrelevant.

• **MINOR**: alts are very keyword-loaded ("Dataset Minecraft — open world, 2.2M videos indexados para visión computacional") which is borderline keyword-stuffing. Google generally rewards descriptive over keyword-dense.

Consider trimming to "Minecraft gameplay dataset preview" type alts.

5.3 Semantic HTML

- `<header role="banner">`, `<main role="main">`, `<footer role="contentinfo">`, `<nav aria-label>` all present. ✓
- `<section aria-labelledby>` consistently used. ✓
- **Carousel buttons are** `<div onclick>` (lines 709, 710, 844–847). **HIGH A11y finding**:

- Not in tab order (no `tabindex`)

- No keyboard handler (Enter/Space don't fire)

- Screen readers announce as generic "group" not "button"

- Fix: replace with `<button type="button" onclick="moveCarousel(-1)" aria-label="Previous slide" class="carousel-btn prev">…</button>`

- `<a href="#">` for footer privacy/licenses (lines 1163–1164) — empty hrefs are valid HTML but UX-broken (see 4.7).

5.4 ARIA / focus management

- **Skip link** (`.skip-link`, line 535) — proper "skip to main content" pattern. ✓
- **Focus-visible** styles (lines 463–490) — strong outline + box-shadow for inputs. ✓
- `prefers-reduced-motion` handled (lines 493–504). ✓
- `aria-expanded="false"` on mobile menu button (line 551) — but I see no JS that toggles this when the menu opens. **LOW finding**: stale ARIA state if the mobile menu is interactive. (Looking at the markup, there's no actual mobile menu drawer — the button is decorative? Verify.)
- **Form**: `aria-labelledby="contacto-title"` + `aria-describedby="contacto-form-help"` ✓ excellent.
- **Honeypot input** (line 1136) has `aria-hidden="true"` ✓
- **Status messages** on form submit (`contact-form.js:138-149`) — the inserted `<p data-status>` doesn't have `role="status"` or `aria-live="polite"`, so screen readers won't announce success/error. **MEDIUM A11y finding**: add `role="status" aria-live="polite"` to that node.

5.5 Color contrast (WCAG)

Quick audit of palette:

- **Body text**: `#9CA3AF` (gray-400) on `#0B0E14` (dark-900) ≈ contrast ratio **6.5:1** — passes WCAG AA for normal text (4.5:1). ✓
- **Muted text**: `#6B7280` (gray-500) on dark-900 ≈ **4.0:1** — fails AA for normal, passes for large text only. Used for footer copyright and card descriptions (e.g., line 977 `text-xs text-gray-500`). **MEDIUM A11y finding**: bump to `gray-400` for accessibility.
- **Placeholder text**: `gray-600` (#4B5563) on dark-900 ≈ **2.3:1** — fails. Placeholders are exempt from strict WCAG (per spec they're not a "primary content"), but visible-on-load placeholders are best at 3:1+.
- **Brand accents**: brand-500 (#14B8A6), accent-cyan (#22D3EE), accent-purple (#A855F7) on dark — all pass 4.5:1.

5.6 Internal linking, anchor text

- **Nav** (line 542–547): 5 anchor links to sections, descriptive labels ("Métricas del Dataset", "Dataset Intelligence", "Juegos F2P", "Juegos Publishers", "Formato de Entrega"). ✓
- **CTA buttons**: "Contáctanos" + "Ver ejemplos" with proper href. ✓
- **No external links** other than image hosts and analytics (auto). Footer privacy/licenses links are placeholders (see 4.7).
- **Rich anchor diversity** is hard on a single-page landing — current setup is fine.

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6. Core Web Vitals — projected

Best-effort projections for a typical mobile Slow-4G visitor (no real run):

Metric	Projected	Target	Notes
LCP	~1.■8–2.■5 s	<2.■5 s	Hero `<h1>` (line 560) is text-■only, no image.■ Painted after Tailwind-■inlined CSS + Inter swap.■ Mostly OK; FA preload steals 156 KB that could go to Inter.
INP	~80–150 ms	<200 ms	Light JS, deferred.■ Carousel onclick is fast.
CLS	<0.■05	<0.■1	Critical CSS prevents FOUC; `width`/■`height` on imgs prevent shift; font-■display:■swap may cause minor swap shift but Inter & system stack are similar metrics.
TTFB	80–250 ms	<800 ms	CloudFront edge in user's region, S3 origin in us-■east-■1.
FCP	~0.■9–1.■4 s	<1.■8 s	Critical CSS inline + minimal blocking.■ ✓

Likely Lighthouse Performance score: **88–94 mobile, 96–99 desktop** (educated estimate, real run advised).

7. Recommendations (prioritized)

Performance

1. **[HIGH | LCP –80–150 ms | LOW effort]** Re-purge Tailwind with `content: ['index.html']` and re-inline. Saves ~15 KB on every cold HTML request.
2. **[MEDIUM | LCP –50–80 ms | LOW]** Drop `<link rel="preload" as="font" … fa-solid-900.woff2>` (line 252). Icons aren't LCP-blocking; the 156 KB preload steals from Inter. Also consider re-subsetting fa-solid to ~30 used glyphs (~15 KB target).
3. **[MEDIUM | TBT –300–500 ms first visit | MEDIUM]** Combine the 6 public bootstrap fetches (`/api/stats`, `/api/games`, `/api/site-links`, `/api/delivery-options`, `/api/info-columns`, `/api/cta-tags`) into one `/api/landing-bootstrap`. Reduces 6 round-trips to 1.
4. **[MEDIUM | initial paint quality | LOW]** Hard-code WebP + PNG `<picture>` directly in `index.html` for game card images, instead of relying on JS upgrade. Saves ~50% image weight on first paint.
5. **[LOW | bandwidth | LOW]** Optimize `og-preview.png` (100 KB → ~40 KB JPG). Doesn't affect page perf, but speeds up social previews.
6. **[LOW | dead code | LOW]** Delete or stop-syncing `assets/css/tailwind.min.css` (25 KB) since it's not referenced.
7. **[LOW | cleanup | LOW]** Strip `fa-brands-400.woff2` and `fa-v4compatibility.woff2` `@font-face` rules from `fa-subset.min.css` — referenced files don't exist.
8. **[LOW | cleanup | LOW]** Remove leftover `<style data-grapesjs-styles="true">` block (lines 520–530). Move the two ID rules into the main stylesheet or inline critical CSS.
9. **[LOW | infra | MEDIUM]** Add fingerprinted asset paths (e.g., `assets/css/fa-subset.7c3a91.min.css`) so CloudFront can cache assets at `max-age=31536000, immutable`. Eliminates the current 1-hour ceiling.

SEO

1. **[HIGH | indexing accuracy | LOW]** Resolve `<html lang="es">` vs JS `currentLang='en'` mismatch. Pick one canonical language for first paint and align `<html lang>` + Twitter Card title (currently English while OG is Spanish).

2. **[MEDIUM | duplicate content | LOW]** Verify and (if needed) configure `www.monoulabs.ai` → `monoulabs.ai` 301 redirect at CloudFront level.

3. **[MEDIUM | metadata integrity | LOW]** Either implement `#privacy` / `#data-licenses` sections or remove their references from llms.txt + schema.org Service.

4. **[MEDIUM | freshness signals | LOW]** Auto-update `dateModified` in JSON-LD Dataset and `lastmod` in sitemap.xml on every deploy (script in deploy.sh). Currently lags by ~4 days.

5. **[LOW | entity graph | LOW]** Populate `sameAs` in Organization JSON-LD with LinkedIn / Twitter / GitHub URLs.

6. **[LOW | snippet length | LOW]** Trim meta description to ~155 chars to avoid SERP truncation.

A11y (also affects SEO indirectly via Lighthouse)

1. **[HIGH | WCAG 2.1.1, 4.1.2 | LOW]** Replace carousel `<div onclick=>` with `<button type="button">` (lines 709, 710, 844–847). Add `aria-label="Previous slide"` / `"Next slide"` / `"Slide N"`. Saves a real keyboard-navigation bug.

2. **[MEDIUM | WCAG 1.4.3 | LOW]** Bump `text-gray-500` (#6B7280) to `text-gray-400` for body-size text (e.g., line 977 game category labels). Fails 4.5:1 contrast.

3. **[MEDIUM | WCAG 4.1.3 | LOW]** Add `role="status" aria-live="polite"` to the form status `<p data-status>` injected by `contact-form.js:140`.

4. **[LOW | heading hierarchy | LOW]** Either demote stat values from `<h3>` to `<p>`, or accept the warning. Lighthouse won't fail you.

5. **[LOW | ARIA hygiene | LOW]** Wire up the mobile menu `aria-expanded` toggle (line 551), or remove the attribute if the menu has no drawer.

6. **[LOW | h2→h4 skip | LOW]** Insert a hidden `<h3 class="sr-only">F2P Game Catalog</h3>` inside the carousel section to maintain heading order.

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8. What's already done well

This list is intentionally long because the v2.4 → v3.1 audits clearly invested in best-practices.

Performance wins

- **Critical CSS inlined** for above-the-fold (lines 215–229).
- **Tailwind compiled statically + inlined** (vs CDN).
- **Inter self-hosted** with `unicode-range: latin` and `font-display: swap` (lines 260–274). Killed `fonts.googleapis.com` 750 ms hop.
- **Inter 400/600/700 preloaded** with correct `crossorigin` (lines 64–66).
- **All `<img>` have `loading="lazy"`, `decoding="async"`, explicit width/height**.
- **FA subset CSS loaded async** via `preload + onload=this.rel='stylesheet'` pattern (line 248).
- **All 7 deferred JS scripts** use `defer` (lines 1314–1321).
- `analytics-loader.js` **uses** `requestIdleCallback` to push GA4/GTM out of LCP critical path (lines 232–243). Top-tier optimization.
- `localStorage` **cache (5 min TTL)** for all 6 public API endpoints. Eliminates round-trips on repeat visits.
- **Honeypot anti-bot field** with proper `tabindex="-1"` and visually-hidden styles (line 1136).
- `prefers-reduced-motion` **respected** (lines 493–504).
- **Skip-link** properly implemented (line 535, .skip-link CSS line 441).
- `<picture>` **+ WebP source pattern** supported in `games-loader.js:97–113` (waiting for backend `image_webp_url`).

SEO wins

- **Comprehensive meta block**: title, description, robots, canonical, viewport, theme-color, color-scheme, format-detection (lines 5–77).
- **Full OG block** with width/height/type/secure_url/locale/alternate/site_name (lines 18–30) — nails LinkedIn/Slack/Facebook previews.

- **Twitter Card** with summary_large_image, alts, site, creator (lines 33–39).
- **Four JSON-LD blocks**: Dataset (rare and high-value for Google Dataset Search), Organization with `@id`, Service referencing Org, WebSite (lines 83–213).
- **Self-referencing hreflang** for `es` + `x-default` (lines 14–15) — explicit and correct given EN is a UX toggle.
- **Sitemap.xml** with hreflang alternates (lines 11–12) and lastmod.
- **robots.txt is exemplary**: explicit allow for AI search bots (Perplexity, OAI-SearchBot, Google-Extended, Applebot-Extended, MistralAI), explicit disallow for AI training crawlers (GPTBot, ClaudeBot, CCBot, etc.), throttle for SEO scrapers, sitemap reference. This is rare and well-thought-out for a B2B AI-data company.
- `llms.txt` present at root, structured for LLM retrieval — modern and forward-looking.
- `<meta name="referrer" content="strict-origin-when-cross-origin">` for privacy.
- **Spanish alt text on all images** with relevant SEO keywords.

A11y wins

- **Skip-to-content link** (line 535).
- **Focus-visible** with strong outlines (lines 463–490).
- `prefers-reduced-motion` respected globally (lines 493–504).
- **Form labels properly associated** via `for`/`id` (lines 1125–1133).
- **Form** `aria-labelledby` **+** `aria-describedby` (line 1123).
- `aria-label` **on icon-only nav** (mobile menu button line 551, social header line 541).
- `<section aria-labelledby>` consistently used.
- **Honeypot field hidden via off-screen positioning** (not `display:none`, which screen readers might still announce).
- `autocomplete` **attributes** on form inputs (`name`, `email`, `organization`).
- `inputmode="email"` for the email field — better mobile keyboard.

Backend / infra wins

- **AWS-native stack with IAM roles** for SES (no API keys to rotate).
- **Rate limiting** on contact form (5/min, 30/hr, 100/day).
- **Honeypot** + email regex + per-field length limits.
- `X-Content-Type-Options: nosniff` **+** `Referrer-Policy` **+** `X-Frame-Options: DENY` on all API responses (`server/app.py:148–155`).
- `Cache-Control: no-store` **on admin endpoints**.
- `force_ipv4()` **defensive context manager** for outbound HTTPS (legacy from Render, kept defensively).
- **Indexes on** `contacts.created_at`**,** `email`**,** `status` (db.py:38–40).

End of report.